

Dung Bui

Postdoctoral Researcher in Cryptography, Northwestern University

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About

My research focuses on cryptography, including secure multiparty computation, zero-knowledge proofs, and post-quantum cryptography.

Positions

Postdoctoral Researcher

Department of Computer Science, Northwestern University
Host: Xiao Wang.

Evanston, IL, USA

July 2026 – present

Postdoctoral Researcher

LIP6, Sorbonne Université / CNRS

Hosts: Eleni Diamanti, Alex B. Grilo, Ludovic Perret.

Paris, France

June 2025 – June 2026

Education

Université Paris Cité / CNRS

PhD in Computer Science, IRIF

Advisor: Geoffroy Couteau.

Paris, France

Sept. 2021 – Mar. 2025

- Thesis: *Efficient Secure Computation from Correlated Pseudorandomness*; defended March 2025.
- Honourable mention, PhD Thesis Prize of GDR Sécurité (CNRS).

Université de Limoges

MSc in Mathematics, Cryptology, Coding & Applications (CRYPTIS)

- Highest honours, ranked 1st; fully funded by the Vingroup Scholarship Programme.

Limoges, France

Sept. 2019 – July 2021

École Normale Supérieure de Hanoi

BSc in Pure Mathematics; highest honours, top 5%

Hanoi, Vietnam

Sept. 2015 – July 2019

Research Experience

Visiting Researcher

NTT Social Informatics Laboratories

Host: Masayuki Abe.

Tokyo, Japan

June – Aug. 2024

- Multi-round zero-knowledge proofs and applications to advanced post-quantum signatures based on MPC.

Visiting Researcher

COSIC, KU Leuven

Host: Nigel Smart.

Leuven, Belgium

July – Aug. 2023

- Efficient post-quantum signatures from multiparty computation.

Research Intern

IRIF, Université Paris Cité
Advisor: Geoffroy Couteau.

Paris, France
Mar. – Aug. 2021

- Efficient MPC protocols from pseudorandom correlation generators.

Research Intern

XLIM, Université de Limoges
Advisor: Duong Hieu Phan.

Limoges, France
June – Aug. 2020

- Private set intersection and its applications.

Publications

Author lists follow the alphabetical convention of cryptography. Full versions on [DBLP](#) or [Google Scholar](#).

2026

1. Practical Adaptor Signatures for NP from Online/Offline NIZK
Masayuki Abe, **Dung Bui**, Kelong Cong, Miyako Ohkubo, Zehua Shang, Akira Takahashi, Mehdi Tibouchi
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2026
2. Critical Round in Multi-Round Proofs: Compositions and Transformation to Trapdoor Commitments
Masayuki Abe, David Balbás, **Dung Bui**, Miyako Ohkubo, Zehua Shang, Akira Takahashi, Mehdi Tibouchi
International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), 2026

2025

1. New Framework for Structure-Aware PSI from Distributed Function Secret Sharing
Dung Bui, Gayathri Garimella, Peihan Miao, Phuoc Van Long Pham
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2025
2. Efficient Fuzzy Labeled PSI from Vector Ring-OLE **(Best Paper Award)**
Dung Bui, Kelong Cong
International Conference on Cryptology and Network Security (CANS), 2025
3. Structured-Seed Local Pseudorandom Generators and Their Applications
Benny Applebaum, **Dung Bui**, Geoffroy Couteau, Nikolas Melissaris
International Conference on Randomization and Computation (RANDOM), 2025
4. Improved All-but-One Vector Commitment with Applications to Post-Quantum Signatures
Dung Bui, Kelong Cong, Cyprien Delpéch de Saint Guilhem
Australasian Conference on Information Security and Privacy (ACISP), 2025
5. Efficient Multi-Instance Vector Commitment and Application to Post-Quantum Signatures
Dung Bui
Australasian Conference on Information Security and Privacy (ACISP), 2025

2024

1. FOLEAGE: $\mathbb{F}_4$ OLE-Based Multi-Party Computation for Boolean Circuits
Maxime Bombar, **Dung Bui**, Geoffroy Couteau, Alain Couvreur, Clément Ducros, Sacha Servan-Schreiber
International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2024

2. Faster Signatures from MPC-in-the-Head
Dung Bui, Eliana Carozza, Geoffroy Couteau, Dahmun Goudarzi, Antoine Joux
 International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT), 2024
3. Fast Public-Key Silent OT and More from Constrained Naor-Reingold
Dung Bui, Geoffroy Couteau, Pierre Meyer, Alain Passelègue, Mahshid Riahinia
 International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT), 2024

2023

1. An Efficient ZK Compiler from SIMD Circuits to General Circuits
Dung Bui, Haotian Chu, Geoffroy Couteau, Xiao Wang, Chenkai Weng, Kang Yang, Yu Yu
 Journal of Cryptology (JoC), 2024
2. Improved Private Set Intersection for Sets with Small Entries
Dung Bui, Geoffroy Couteau
 International Conference on Practice and Theory of Public-Key Cryptography (PKC), 2023

Manuscripts

1. Generic Post-Quantum Anamorphic Signatures: Private, Broadcast, and Threshold Channels from Multi-Round Proofs
Dung Bui, Francesco Migliaro
 In submission
2. Critical-Round Special Soundness for Multi-Round Proofs
 Masayuki Abe, David Balbás, **Dung Bui**, Miyako Ohkubo, Zehua Shang, Akira Takahashi, Mehdi Tibouchi
 In submission; Cryptology ePrint Archive, Report 2026/1679

Teaching & Advising

Teaching Assistant and Co-instructor Evanston, IL, USA
 Northwestern University Fall 2026

- COMP_SCI 307: Introduction to Cryptography (undergraduate).

Research Supervision Evanston, IL, USA
 Fengke Li (BSc student, Shanghai Jiao Tong University) July – Dec. 2026
 Co-supervised with Xiao Wang.

- Research internship at Northwestern University.

Talks

Talks are grouped by research theme; venues are listed most recent first.

Private set intersection

Efficient Fuzzy Labeled PSI from Vector Ring-OLE

- CANS 2025, Osaka, Nov. 2025

Efficient PSI from Pseudorandom Correlation Generators

- Crypto Seminar, COSIC (KU Leuven), July 2023
- Vietnam Institute for Advanced Study in Mathematics (VIASM), May 2023 (virtual)

- Algorithms and Complexity seminar, IRIF, Paris, Apr. 2023
- Crypto Student Seminar, CWI Amsterdam, July 2022 (virtual)

Improved Private Set Intersection with Small Entries

- PKC 2023, Atlanta, May 2023

Private Set Intersection from Correlated Randomness

- Journées C2 (Codage & Cryptographie) workshop, Hendaye, Apr. 2022

Post-quantum signatures

Adaptor Signature from VOLEith

- Advanced Seminar on Security and Privacy, University of Illinois Urbana-Champaign (UIUC), Nov. 2026

Optimized MPC-in-the-Head Based Signatures from Puncturable PRF

- Crypto Seminar, NTT Social Informatics Laboratories, Tokyo, Nov. 2025
- Vietnam Institute for Advanced Study in Mathematics (VIASM), Apr. 2025 (virtual)
- Crypto Student Seminar, CWI Amsterdam, Dec. 2024
- Crypto Day, Télécom Paris (Institut Polytechnique de Paris), Nov. 2024

Improved All-but-One Vector Commitment with Applications to Post-Quantum Signatures

- ACISP 2025, Wollongong, July 2025

Faster Signatures from MPC-in-the-Head

- ASIACRYPT 2024, Kolkata, Dec. 2024

MPC in the preprocessing model

FOLEAGE: $\mathbb{F}_4$ OLE-Based Multi-Party Computation for Boolean Circuits

- Crypto Café, Florida Atlantic University, Sept. 2025 (virtual)
- ALMASTY seminar, LIP6, Sorbonne Université, Mar. 2025

Fast Public-Key Silent OT and More from Constrained Naor-Reingold

- AlgoCRYPT seminar, J.P. Morgan, May 2024 (virtual)
- Non-permanents' seminar, IRIF, Paris, May 2024

Efficient MPC from Correlated Randomness

- Crypto Seminar, NTT Social Informatics Laboratories, Tokyo, June 2024

Zero-knowledge proofs

Critical Round in Multi-Round Zero-Knowledge Proofs

- Vietnam Institute for Advanced Study in Mathematics (VIASM), May 2026 (virtual)
- ALMASTY seminar, Sorbonne Université, Apr. 2026
- Cybersécurité DVRC/ESILV workshop, Paris, Mar. 2026
- INRIA C2 (Codage & Cryptographie) seminar, Paris, Jan. 2026

Fellowships & Awards

1. Honourable Mention, 2025 PhD Thesis Prize of GDR Sécurité (CNRS), awarded 2026: annual national award recognizing excellent PhD theses in information security in France.
2. Best Paper Award, CANS 2025, for *Efficient Fuzzy Labeled PSI from Vector Ring-OLE*.
3. Fully funded doctoral grant (36 months), DIM Math Innov / FSMP, 2021–2025: awarded by the Île-de-France Region; up to 9 PhD fellowships across mathematics and computer science.

4. Silver Medal, NSUCRYPTO International Olympiad in Cryptography, 2021.
5. Fully funded scholarship, Vingroup Scholarship Programme (30 000 EUR), 2019–2021, for the CRYPTIS MSc at the Université de Limoges.
6. VIASM Scholarship (3 000 EUR), National Programme for the Development of Mathematics, Vietnam Institute for Advanced Study in Mathematics, 2017–2019.
7. Bronze Medal, Vietnam National Mathematical Olympiad (VMO), 2015: part of the selection process for the International Mathematical Olympiad (IMO).

Professional Service

Programme committee member: CSF 2027; APKC 2026, CCS 2026, ASIACRYPT Artifacts 2026; APKC 2025, LATINCRYPT 2025.

External reviewer: CRYPTO 2025 ($\times 3$), 2026 ($\times 3$); EUROCRYPT 2025, 2026 ($\times 2$); ASIACRYPT 2025, 2026 ($\times 2$); PKC 2026; ICALP 2026; CCS 2025; ACNS 2025; PriCrypt 2025; TCC, CSF and IEEE TIFS, 2022–2024.

Organiser: IRIF Crypto Seminar, 2024–2025.

Outreach

- Member, Advanced Cryptography Research Lab, a joint initiative between VIASM and the Vietnam Government Information Security Commission, 2025– (<https://acr-lab.github.io/>).
- Featured interview, *Meet the Women Experts in Quantum Science*, QICS, Sorbonne Université, 2025: on career paths and motivation in post-quantum cryptography ([link](#)).
- Co-organizer of QATS, an educational board game on quantum mechanics, and activity leader at *Quantum Day (Lumière sur la quantique)*, organized by DIM QuanTiP, PCQT, Quantum Saclay, Atouts Sciences, QICS, and la SFP, 2026 (<https://quantip.org/espace-grand-public/quantum-day/>).
- Featured interview, *Tuổi Trẻ* (Vietnamese national magazine), 2019: on career paths and motivation in cryptography ([link](#)).
- Founder and author of the Vcryptis blog: practical advice on student life and academics at the Université de Limoges, 2019–2021 (<http://vcryptis.tech.blog/>).

Languages

Vietnamese (native) · English (professional proficiency) · French (DELFI B2).

References

Geoffroy Couteau

PhD advisor

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Xiao Wang

Postdoctoral host

Northwestern University

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Masayuki Abe

Research collaborator and host

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